Intro to Metaheuristics

Teamwork 2

TSP problem

Team Members

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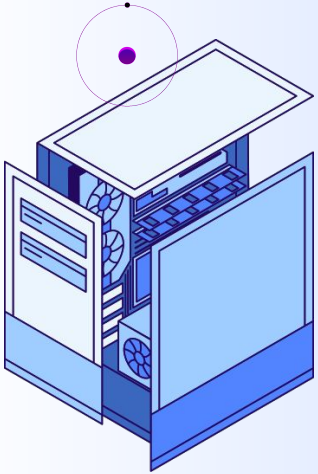
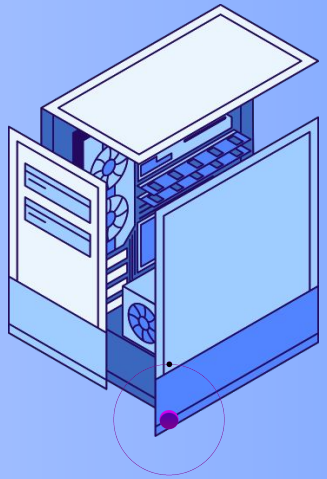
Content

- Introduction to TSP
- Genetic Algorithm(GA).
- Ant Colony Optimization(ACO).
- Hyper Parameter Optimization of GA and ACO.

Introduction to TSP

Traveling Salesman Problem (TSP)

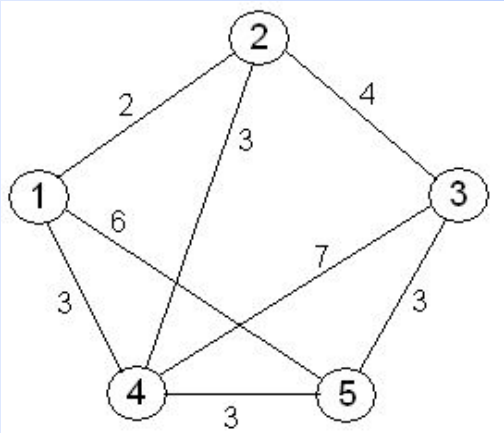




What is TSP? :

TSP is a problem that seeks to find the route with the minimum total distance traveled (or cost) among all possible routes that visit each of multiple cities exactly once and return to the starting point.

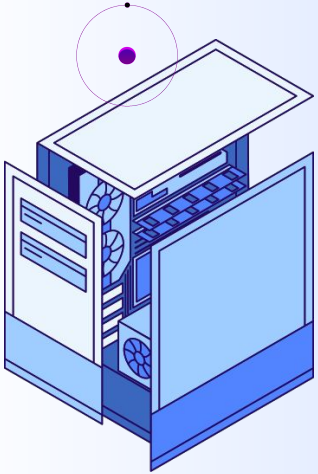
The method of obtaining the optimal solution by calculating all paths has a time complexity of O(n!) and is NP-hard

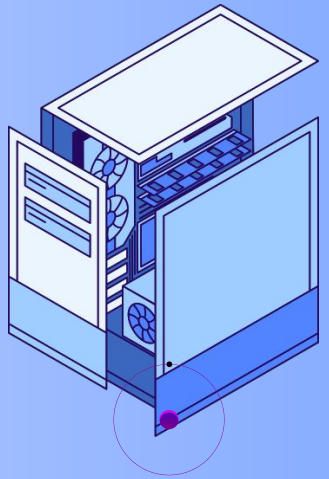


https://commons.wikimedia.org/wiki/File:Example_The_travelling_salesman_problem_(TSP).gif

Genetic Algorithm (GA)

Genetic Algorithm (GA)





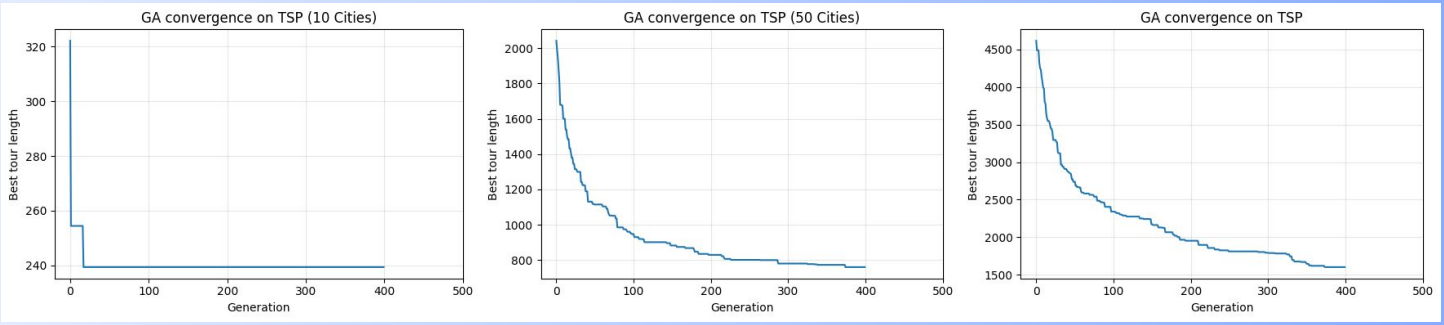
Algorithm Concept:

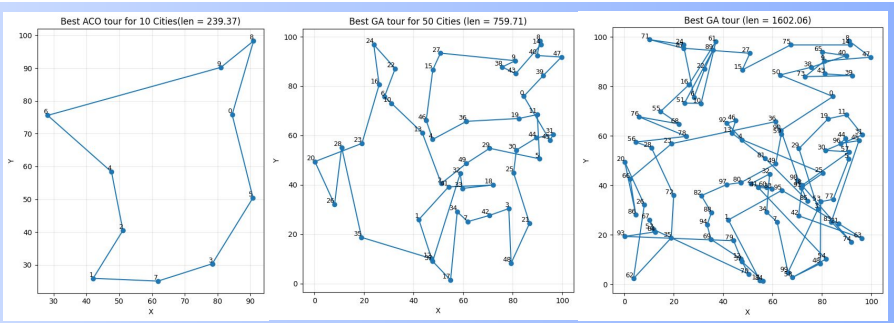
1. Generate initial solutions.
2. Evaluate each cost of solutions and crossover (※ used order crossover)
3. Adaptively mutate solutions randomly

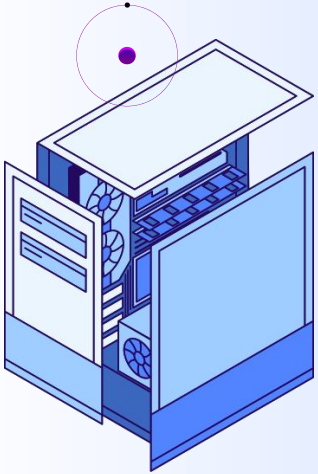
Parameter :

generations	400
pop_size	80
cx_rate	0.9
mut_rate	0.2
elite_size	2
tournament_k	3

GA convergence





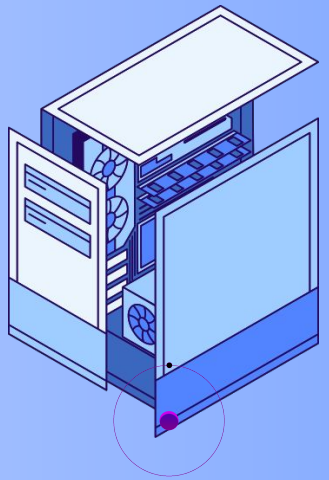


Discussion:

The more cities there are, the slower the approach to the optimal solution becomes.
In 400 generations, there were cases where the optimal solution could not be reached.

Ant Colony Optimization (ACO)

Ant Colony Optimization (ACO)



Goal:

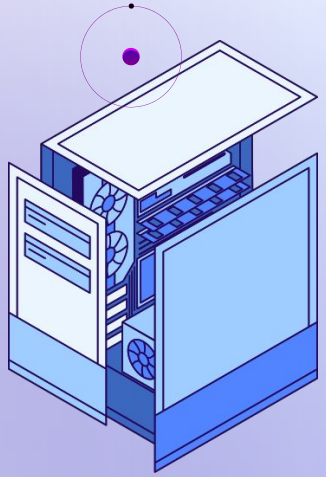
- Find the shortest route visiting all cities once (TSP).

Key Concept:

- ACO simulates how ants find the shortest path between food and the nest using pheromone trails.

Core Mechanism:

- Each “ant” builds a solution (tour).
- Paths with stronger pheromone trails are more likely to be chosen.
- Pheromone intensity increases for shorter tours (reinforcement).
- Evaporation prevents premature convergence.



Algorithm Workflow

Initialization

- Create distance and pheromone matrices (τ_{ij}).
- Set heuristic information ($\eta_{ij} = 1/d_{ij}$).

Construct Solutions

- Each ant incrementally builds a tour based on combined influence of pheromone (τ_{ij}^{α}) and heuristic (η_{ij}^{β}).

Update Pheromones

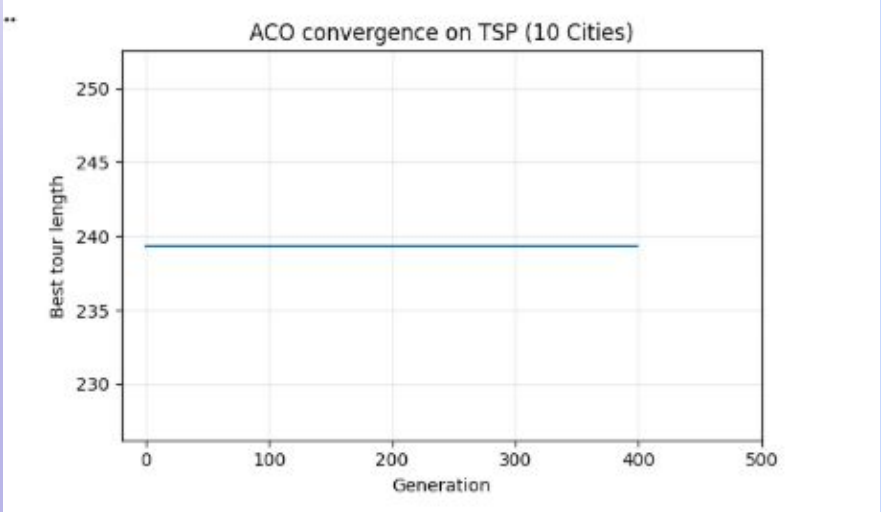
- Evaporate: $\tau_{ij} \leftarrow (1 - \rho) * \tau_{ij}$
- Reinforce: $\tau_{ij} \leftarrow \tau_{ij} + Q/L$, where L = tour length.

Repeat for several iterations.

Parameter Settings

Parameter	Description	Typical Values
α	Pheromone importance	1.0
β	Heuristic importance	5.0
ρ	Evaporation rate	0.1
Q	Pheromone deposit constant	100

Ant Colony Optimization (ACO) - Convergence Example (10 Cities)



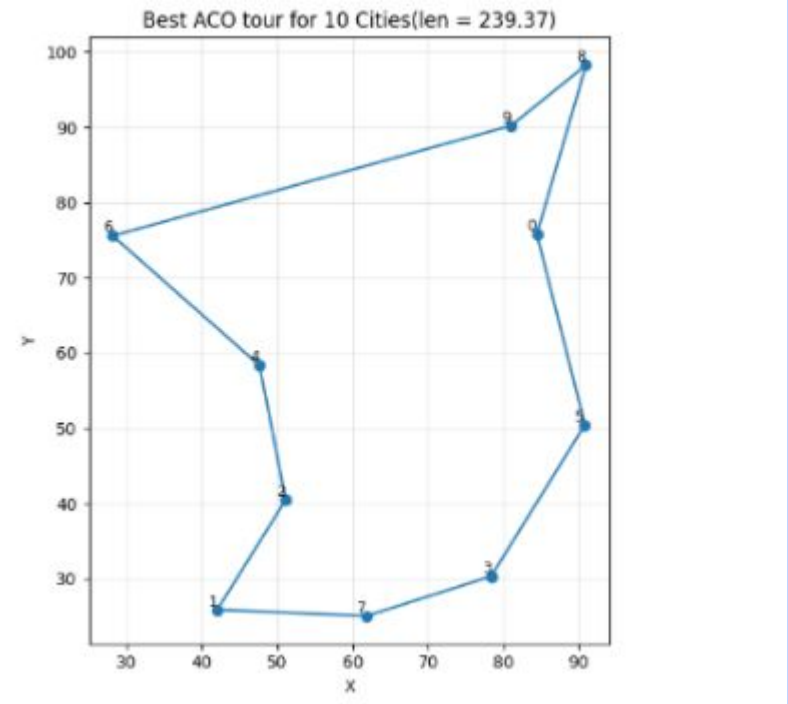
Initial phase (0–100 iterations): Rapid improvement as pheromone trails reinforce shorter routes found by ants.

Later phase: Convergence, the colony collectively focuses on near-optimal paths, and further improvement slows.

Interpretation: Demonstrates the **balance between exploration and exploitation** in ACO; pheromone update drives convergence while random choice prevents premature stagnation.

Result: Best tour length stabilizes around ~240 units, showing effective search but limited variation once equilibrium is reached.

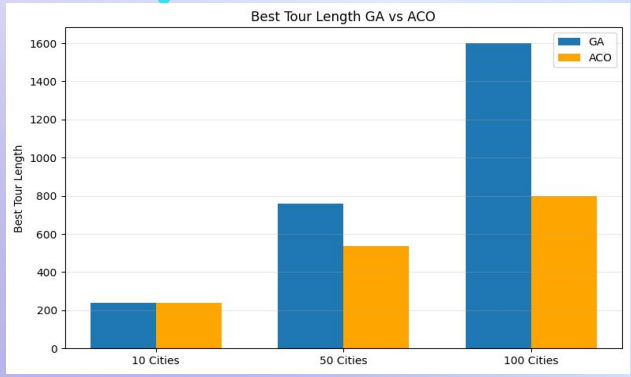
ACO Result: Best Tour (10 Cities)

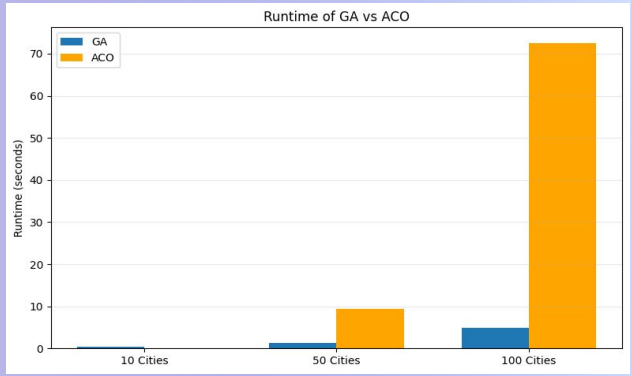


- The figure visualizes the shortest path found after convergence (tour length ≈ 239.37).
- Each point represents a city; the path shows the optimized visiting sequence.
- The near-convex loop pattern suggests ACO effectively learned spatial structure and avoided redundant crossings.

Experimental Results

Experimental Results

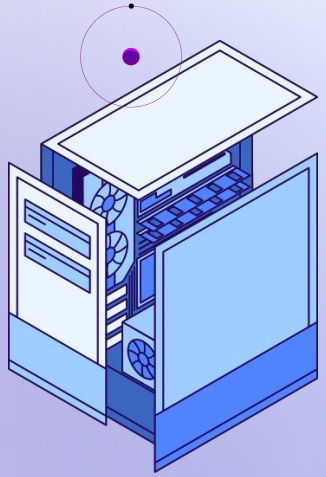


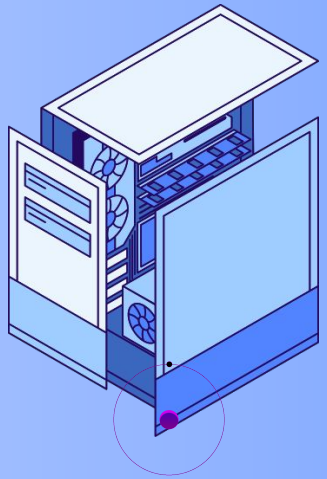


	Algorithm	Best Cost	Runtime (sec)
0	GA (10)	239.372468	0.338811
1	ACO (10)	239.372468	0.089553
2	GA (50)	759.713257	1.259098
3	ACO (50)	536.256224	9.420384
4	GA (100)	1602.059123	4.913047
5	ACO(100)	797.454289	72.603525

Hyper Parameter Optimization

Approach for Optimization





Goal:

- Primary goal is to achieve a better final cost for the tour.
- Secondary goal is to improve runtime while making sure the current best cost is maintained.

Optimized Parameters:

- **GA:** Population size, Crossover rate, mutation rate, elitism, tournament k, number of generations(optimized only in time optimization phase).
- **ACO:** Number of ants, alpha, beta, rho, Q, iterations(optimized only in the time optimization phase)

Constraints:

- Iteration or Generation budget is 400 in both GA and ACO.
- Trial limit for a study in Hyper parameter optimization is 120.

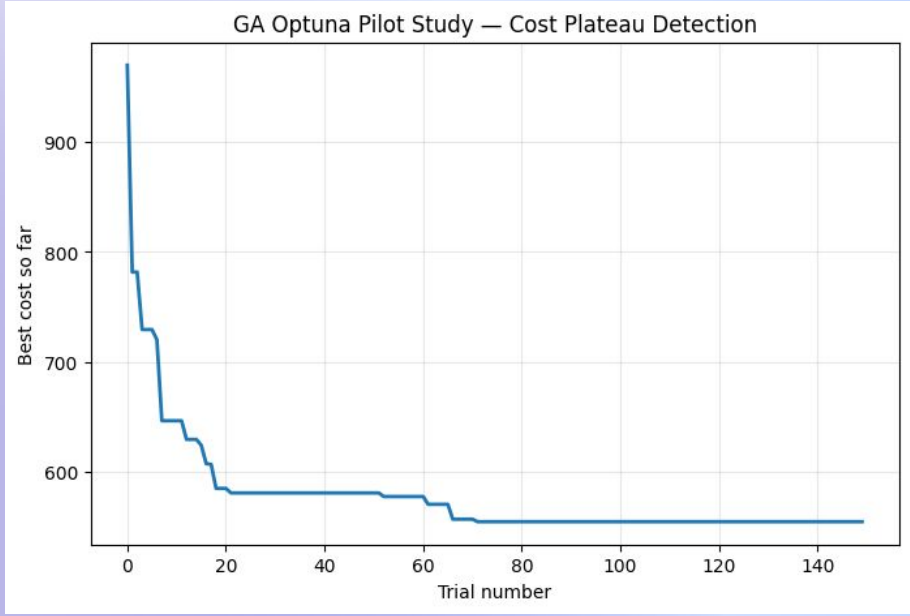
Optimization Workflow

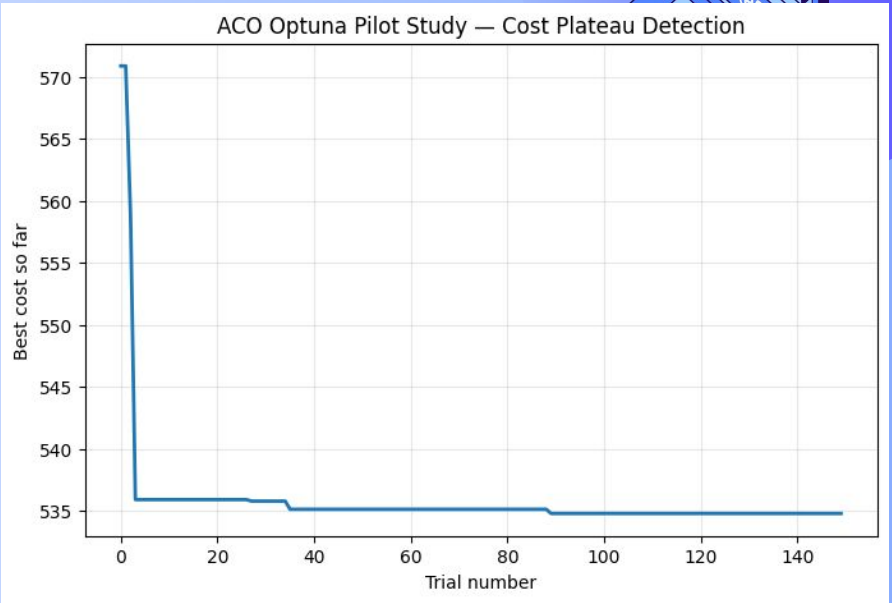
- In optimization of both the algorithms, there are two phases each for cost optimization and time optimization.
- In GA, cost is optimized for 75 trials and time is optimized for remaining 45 trials.
- In ACO, cost is optimized for 40 trials, and time is optimized for remaining 80 trials.

Pilot Study of Cost Optimization

- In optimization of both the algorithms, there are two phases each for cost optimization and time optimization.
- In GA, cost is optimized for 75 trials and time is optimized for remaining 45 trials.
- In ACO, cost is optimized for 40 trials, and time is optimized for remaining 80 trials.

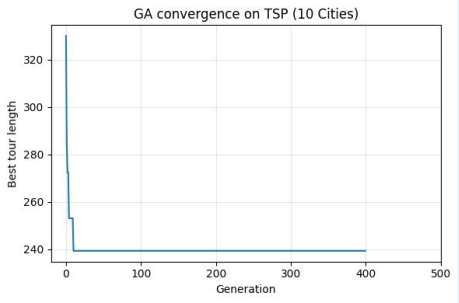
Pilot Study

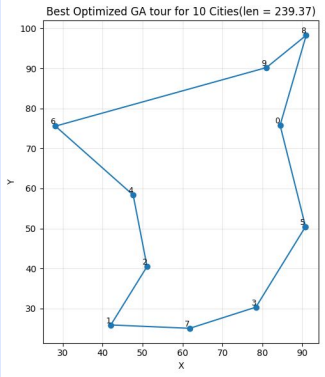

GA Optuna Pilot Study — Cost Plateau Detection
Best cost so far
900
800
700
600
0
20
40
60
80
100
120
140
Trial number

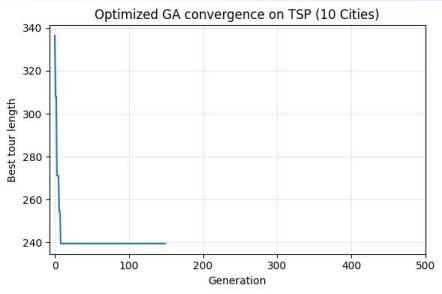

ACO Optuna Pilot Study — Cost Plateau Detection
Best cost so far
570
565
560
555
550
545
540
535
0
20
40
60
80
100
120
140
Trial number

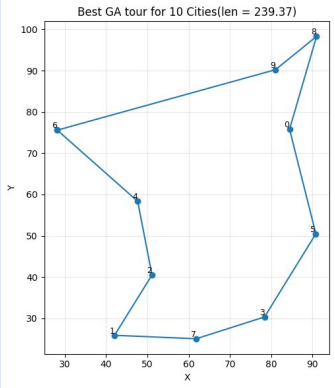
Optimization: 10 Cities

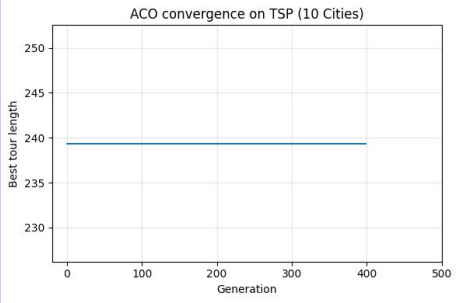
Original and Optimized Performance Comparison(1)

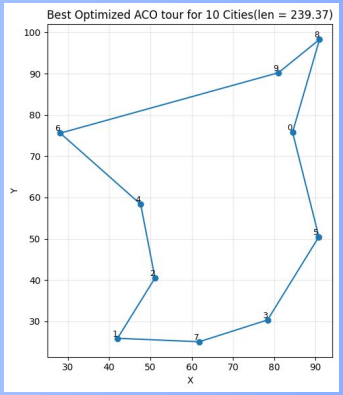


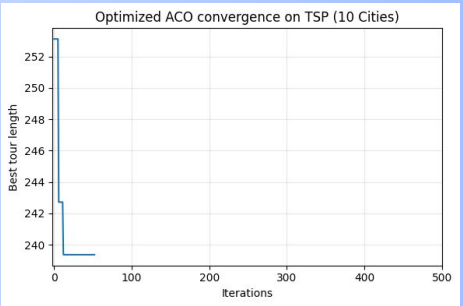


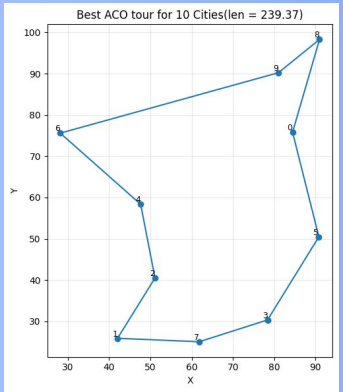




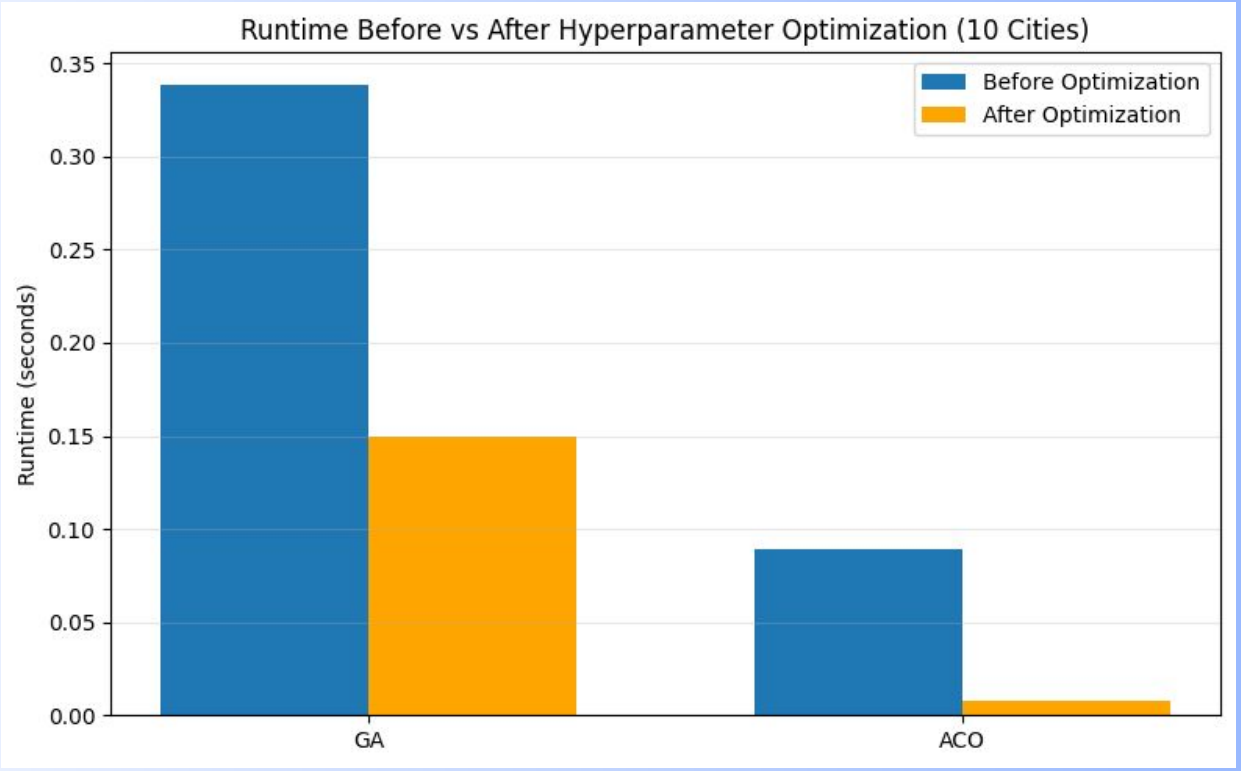






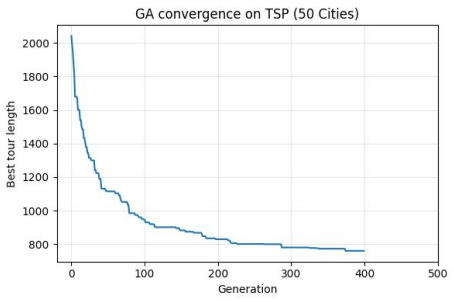


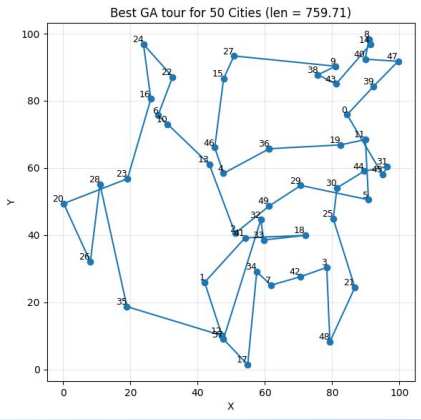
Original and Optimized Performance Comparison(2)

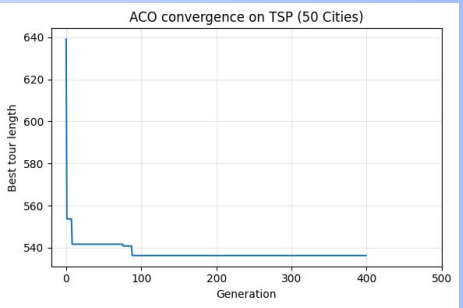


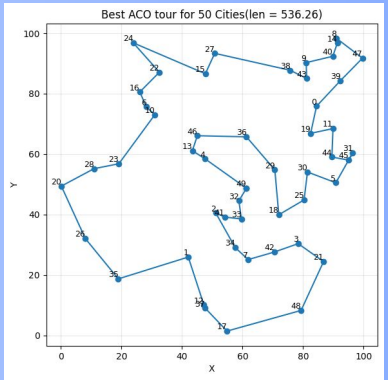
Optimization: 50 Cities

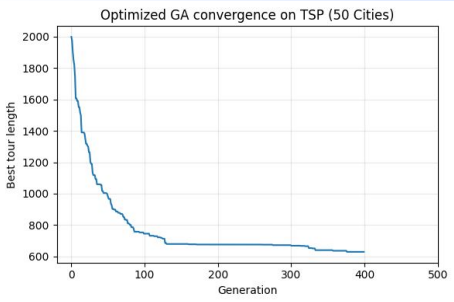
Original and Optimized Performance Comparison(1)

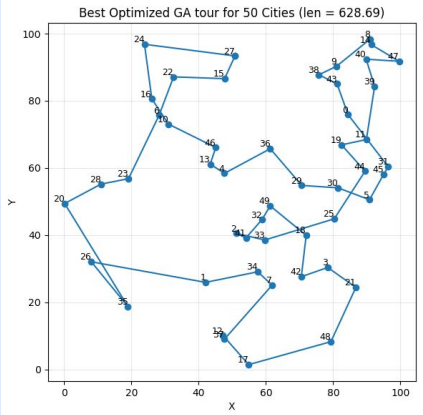


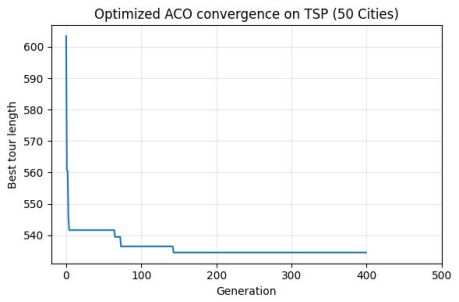


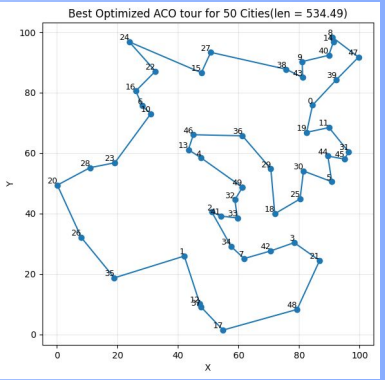












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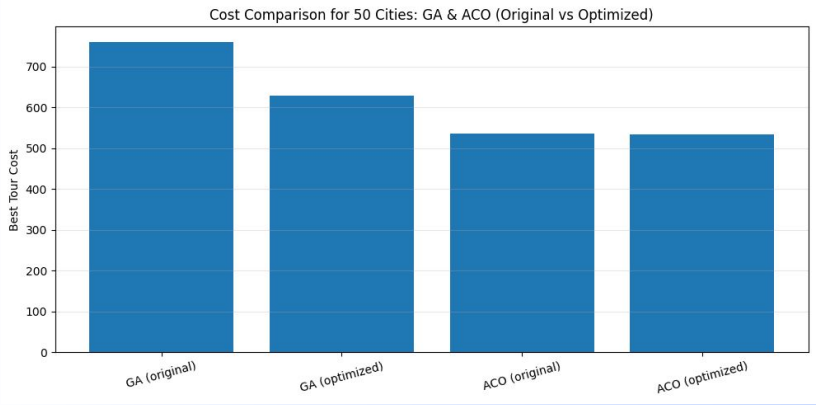
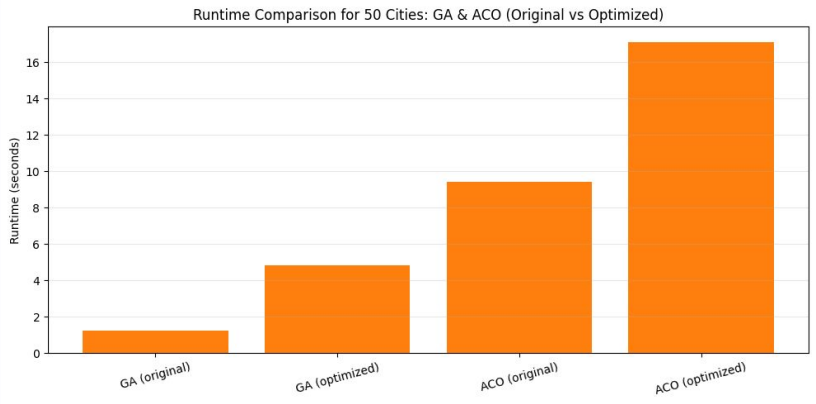
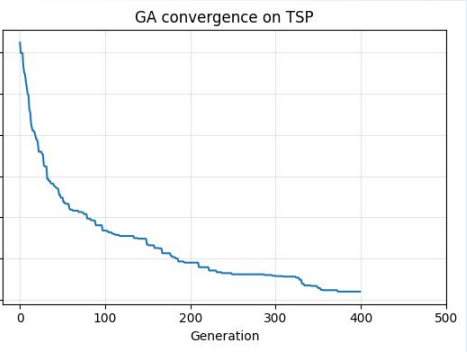


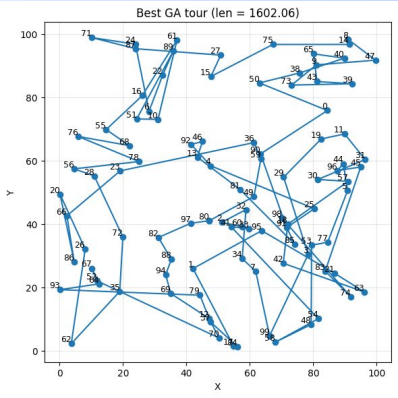


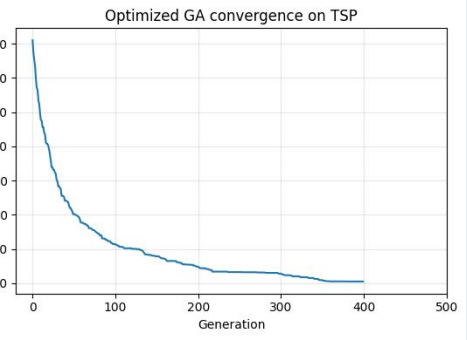
Algorithm	Best Cost	Runtime (sec)
GA (Original)	759.713257	1.259098
GA (Optimized)	628.686094	4.834443
ACO (Original)	536.256224	9.420384
ACO (Optimized)	534.494447	17.105600

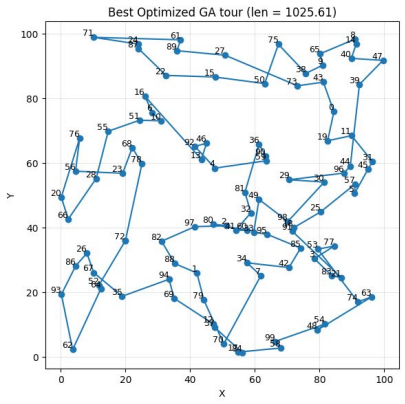
Optimization: 100 Cities

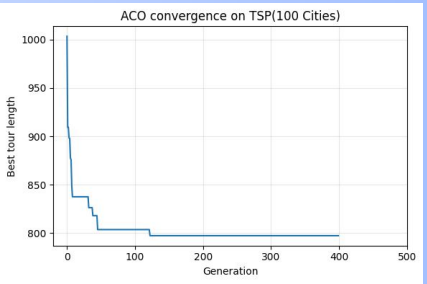
Original and Optimized Performance Comparison(1)

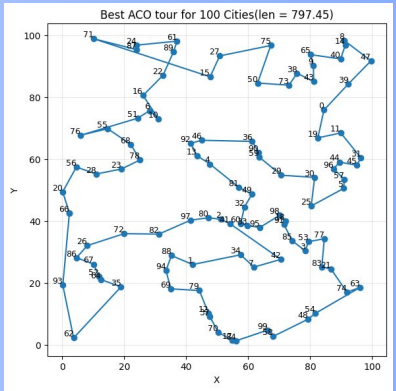


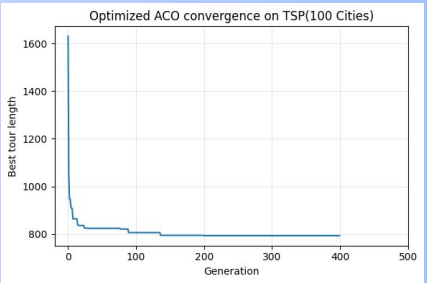


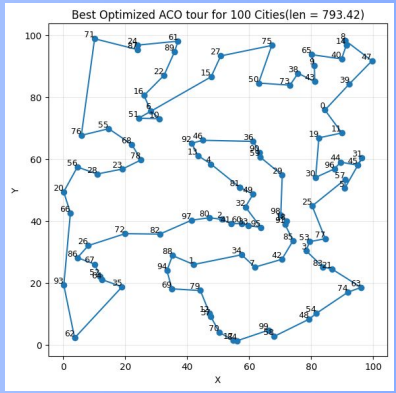




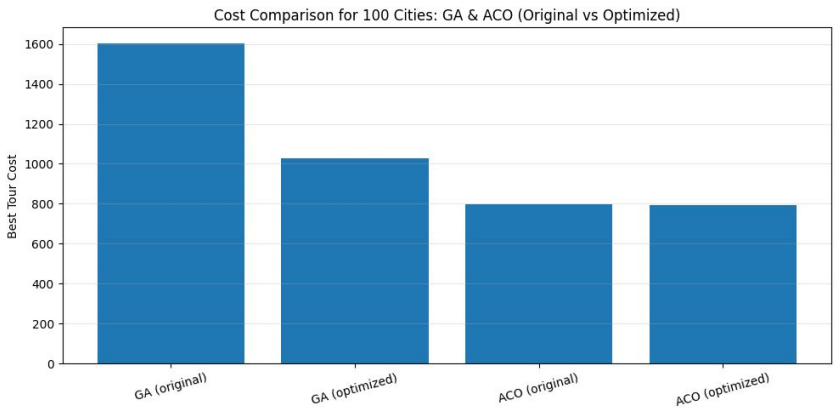


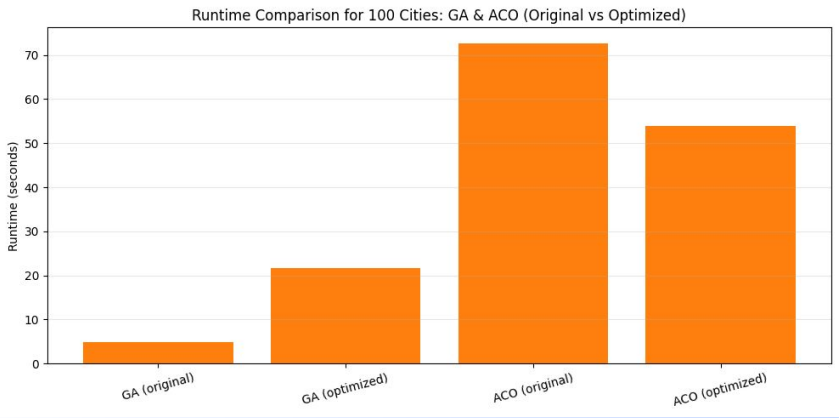






Original and Optimized Performance Comparison(2)





Algorithm	Best Cost	Runtime (sec)
GA (Original)	1602.059123	4.913047
GA (Optimized)	1025.609266	21.691031
ACO (Original)	797.454289	72.603525
ACO (Optimized)	793.415001	53.980039

Contribution

- **Kohei Kuwana:** Apply GA to the TSP Problem: Includes coding, testing, Creating Presentation slides, Reviewing Report and Presentation slides.

- **Minnah Amir:** Apply ACO to TSP problem: Includes coding, testing, Creating Presentation Slides, Reviewing Report and Presentation slides.

- **Tithira Withanaarachchi:** Coordinating work, Parameter Optimization of the application of GA and ACO algorithms to TSP Problem: Coding, testing, Documenting with explanations, Creating Visualizations of the results, Creating Presentation slides, Reviewing Report and Presentation slides.

THANK YOU!

Feel Free to ask any questions…..